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INTERDISCIPLINARY PROJECT REPORT

Operational State Recognition and Braking Behaviour Analysis for Predictive Maintenance on Metro Trains

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1 Introduction

Much rail maintenance still runs on fixed intervals: a component is inspected or replaced after a set mileage or time, regardless of how hard it has actually been used. That wastes healthy parts and still misses parts that fail early under unusual load. Predictive maintenance ties each intervention to the measured condition of the equipment instead, promising less wasted material and fewer unnecessary replacements. The friction brakes of a metro train are a good place to try this: the train brakes at almost every stop, hundreds of times a day, so the brake components are cycled constantly and any degradation should leave a large, repeating signal. This report analyses the MetroAT dataset [7]: one year of 1 Hz recordings from a single Wiener Linien metro train, covering the pneumatic system together with train speed and mode signals, and annotated with documented failure and maintenance events. All sensor values are min-max normalised, so the analysis works in relative rather than physical units. Following the project proposal, the work runs in two phases along the CRISP-DM cycle [8]. The first recognises the train’s operational regimes without supervision, clustering the motion signal into a small set of interpretable states. The second examines the braking that dominates mechanical wear: individual deceleration events are extracted, their intensity characterised, and braking behaviour tested for any trend ahead of documented brake-system failures.

Four questions run through the report: (1) whether distinct operational regimes emerge from the motion data alone, without fixing their number in advance; (2) given those regimes, whether the air-system sensors can recover the deceleration regime using only signals that did not help define it; (3) whether deceleration intensity falls into discrete classes or forms a continuum, and how much of it the air-system sensors can explain; and (4) whether braking behaviour carries any detectable signal ahead of documented brake-system failures. The remainder of the report is organised by topic rather than by project phase.

Structure. Chapter 2 describes how the raw recordings become the analysis units and how the sensors are partitioned into feature roles. Chapter 3 derives and validates four motion-only operational states. Chapter 4 tests whether the air-system sensors alone can recover the deceleration state. Chapter 5 extracts individual deceleration events and regresses their intensity and dissipated energy from the air sensors. Chapter 6 examines the link to documented failures and the maintenance calendar and reports a near-null pre-failure result. Chapter 7 draws the findings together and points to what is worth doing next.

Scope. The conclusions are bounded by the data: a single train, an eight-and-a-half-month training window, and roughly ten documented failures, eight of them brake-system failures; the study is exploratory and not meant as a deployable failure detector. Differences between individual wagons are not analysed: the many near-duplicate per-wagon channels are instead collapsed by principal component analysis during preprocessing (Chapter 2, Figure 2.1).

2 Preprocessing

This chapter describes how the raw MetroAT recordings became the units of analysis, and the design choices along the way that affect every later result.

2.1 The dataset

MetroAT [7] is delivered as Parquet, so no conversion from CSV was needed. The schema has 109 columns: 74 continuous sensor channels, 20 binary channels, 7 operational-mode signals, 4 failure and maintenance markers, and 4 timestamp columns. The recordings run at 1 Hz over roughly one year.

Several properties of the data shaped the processing:

- **Normalised values.** All sensor channels are min-max normalised to approximately $[0, 1]$; they are not physical units. Every threshold used in this study is therefore empirical or quantile-based, and no number in this report should be read as bar, ms^{-2} , or kPa.
- **Unsorted rows.** Rows are not guaranteed to be time-sorted within a daily file, so each file is sorted on load before any derivative is computed.
- **Streaming.** The data is processed one daily file at a time, keeping peak memory below 4 GB rather than loading the full year at once.
- **Per-wagon replication.** Many pneumatic channels are duplicated across wagons, producing 214 sensor pairs with $|r| > 0.98$. No columns are deleted by hand. Instead, where a compact input is needed, principal component analysis is applied to the pneumatic set and the leading components are used in place of the raw channels; because so many channels are near-duplicates, a handful of components already captures most of the variance (Figure 2.1).

2.2 Train and held-out split

Models are fitted on a training window (June 2024 to 11 February 2025, about 14.3 million rows) and scored on a held-out window from 12 February to June 2025 that the models never see during fitting. Every out-of-sample figure reported later therefore comes from the held-out window. The training window contains 10 documented failures and 11 maintenance events, 8 of the failures being brake-system failures; these small counts are what make the pre-failure analysis of Chapter 6 exploratory rather than conclusive.

2.3 Windowing

The unit of analysis is a non-overlapping ten-second window. The roughly 30 million raw rows reduce to about 1.49 million windows. For each window, summary statistics are computed per channel (means and standard deviations, and minima and maxima for the kinematic channels), so that each window carries a compact description of both motion and air-system behaviour over its ten seconds. The ten-second choice is a compromise between resolution and stability; its robustness is examined directly in Section 3.4.

2.4 Derived kinematics

Train speed is the only raw motion channel. From it, acceleration is derived as the rate of change of speed and jerk as the rate of change of acceleration. These derived channels, summarised per window, are the sole basis for the operational states of Chapter 3. Keeping the state definition to motion alone is the concrete consequence of the feature-role split introduced in Section 2.5.

2.5 Feature roles

This project runs clustering, classification, and regression over the *same* set of MetroAT columns: the operational states of Chapter 3 are clustered from the data, and later chapters predict those states and their braking intensity from it again. Whenever the same columns can both define a target and be offered as predictors, leakage becomes a real risk: a predictor that helped define the very thing it is asked to predict will score near-perfectly and reveal nothing. To avoid this, every feature is assigned a *role* in a ledger (`feature_roles.csv`) that fixes whether it may serve as a predictor. The columns are split as follows:

- **velocity:** the motion channels (speed, acceleration, jerk, and the derived deceleration-energy axis). These *define* the operational states and the braking targets, so they are used only as targets and never as predictors.
- **actuation:** the brake command and its immediate pneumatic response (brake-cylinder pressure, proportional-valve pressure, spring-brake pressure, pneumatic braking force). Predictor-eligible, but closely tied to the brake demand.
- **auxiliary:** independent health channels that do not carry the brake command directly (main-reservoir pressure, load pressure, load signal, energy-braking-resistance, ambient temperature, compressor-running). Predictor-eligible, and the cleanest source of independent signal.
- **other:** timestamps, operating-mode flags, and failure/maintenance markers, used for the train/held-out split, time alignment, and validation, but never as predictors.

Table 2.1 summarises the ledger by role. The distinction between *actuation* and *auxiliary* matters most in Chapter 4, where the two are tested as separate predictor tiers: because the

auxiliary channels never touched the brake command, their ability to recover the deceleration state is the cleanest test that the air system carries genuinely independent signal. The complete assignment of channels to roles is listed in Appendix A.

Table 2.1: Sensor roles from the role ledger, aggregated across the window-level and event-level feature sets. Only *actuation* and *auxiliary* features are predictor-eligible; *velocity* features define the targets and are never used as predictors.

Role	Predictor-eligible	Example channels
velocity	no	train speed, acceleration, jerk, deceleration energy
actuation	yes	brake-cylinder / proportional-valve / spring-brake pressure, pneumatic braking force
auxiliary	yes	main-reservoir pressure, load pressure, load signal, energy-braking-resistance, ambient temperature, compressor
other	no	timestamps, operating-mode flags, failure / maintenance markers

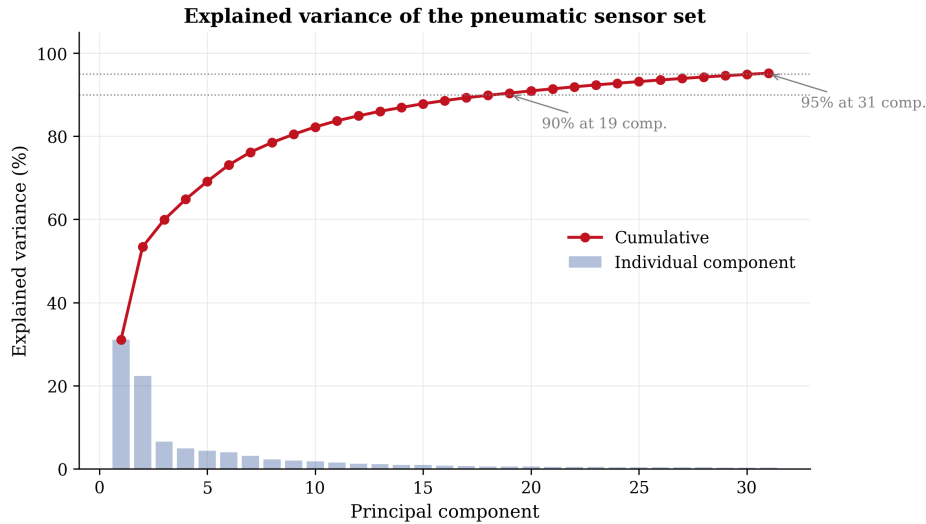


Figure 2.1: Cumulative explained variance of the pneumatic sensor set under PCA. Because 214 sensor pairs are near-duplicates ($|r| > 0.98$) from per-wagon replication, a small number of components captures most of the variation, which is why the redundancy can be absorbed by PCA without dropping any columns.

3 Operational states

This chapter derives the operational regimes of the train and validates them. The regimes are recovered by clustering, and, in line with the feature-role split (Section 2.5), they are defined from motion alone. Only the kinematic window summaries (speed, acceleration, jerk, and their spreads) enter the clustering; the pneumatic sensors are deliberately withheld so that they remain independent evidence in the chapters that follow.

3.1 Clustering into operational states

The states were obtained by k-means clustering of the motion window features, with the number of clusters chosen by internal validation rather than fixed in advance. Two clustering methods (k-means and hierarchical clustering on a sample) were run for k from 2 to 6, and four indicators were compared (Figure 3.1). The *silhouette* score [5] measures how well each window fits its own cluster relative to the nearest alternative; it ranges over $[-1, 1]$, where values near 1 indicate well-separated clusters, values near 0 an ambiguous assignment, and higher is better. The *Davies–Bouldin* index [2] measures the average overlap between a cluster and its most similar neighbour; it is bounded below by 0 and unbounded above, with lower values better. The *Bayesian information criterion* (BIC) [6] of a Gaussian mixture balances goodness of fit against model complexity; it is not bounded to a fixed range and its absolute value is not interpretable, so only differences between models are read, with lower values better. The *adjusted Rand index* (ARI) measures agreement between two clusterings; it takes the value 1 for identical partitions and about 0 for the agreement expected by chance (it can be slightly negative), and is used here both to compare the k-means and hierarchical labels and to assess stability under resampling.

The per- k values of all four indicators are collected in Table 3.1. The silhouette score (Figure 3.1a) rises monotonically with k , from 0.48 at $k = 2$ to 0.60 at $k = 6$, and the Davies–Bouldin index (Figure 3.1b) likewise keeps improving as clusters are added, from 1.20 at $k = 3$ to 0.95 at $k = 6$. Both therefore reward ever finer partitions and neither picks out a value on its own; this is the expected behaviour of internal compactness indices, so the choice cannot rest on them alone. The two clustering methods agree closely at both of the small k that were cross-checked (cross-method ARI 0.81 at $k = 3$ and 0.77 at $k = 4$), and the $k = 4$ solution is stable under resampling (bootstrap ARI about 0.87).

The BIC (Figure 3.1c) is informative for a specific reason: unlike the silhouette and Davies–Bouldin scores it explicitly penalises model complexity, so rather than improving without bound it is expected to flatten once additional clusters stop describing real structure. Its absolute value is not meaningful; only the size of each successive drop is. The largest single drop is the $k = 3 \rightarrow 4$ step ($-121,900$), but the $k = 4 \rightarrow 5$ step is nearly as large ($-110,900$), so the BIC does not by itself separate four clusters from five. What it does show is where the

marginal gain collapses: the $k = 5 \rightarrow 6$ step ($-54,000$) is less than half of either earlier step. The BIC therefore rules out large k but leaves the choice between four and five to be settled on other grounds.

That choice is made on parsimony and physical interpretability. None of the internal indices favours five clusters over four: the silhouette and Davies–Bouldin scores keep rising for any k , and the BIC does not distinguish the two. The smaller model is therefore preferred unless a fifth cluster earns its place with a distinct operating mode, and it does not. Four clusters already exhaust the stop–accelerate–cruise–brake cycle of a metro duty cycle (Table 3.2), each with its own signature in speed and the sign of acceleration, whereas a fifth group adds no regime that is not already present. Four clusters is kept as the most parsimonious description consistent with the metrics and with the physics of the duty cycle.

Table 3.1: Cluster-count selection metrics per k . Silhouette (higher better, range $[-1, 1]$), Davies–Bouldin (lower better, ≥ 0), cross-method ARI (k -means versus hierarchical; 1 is identical, computed for the two small k that were cross-checked), and the Gaussian-mixture BIC (lower better; only differences are meaningful). ΔBIC is the drop from the previous k . The largest drop is at $k = 3 \rightarrow 4$, but $k = 4 \rightarrow 5$ is comparable; the gain collapses only after $k = 5$.

k	Silhouette	Davies–Bouldin	Cross-method ARI	BIC ($\times 10^3$)	ΔBIC ($\times 10^3$)
2	0.476	0.999	n/a	-999.1	n/a
3	0.535	1.196	0.81	-1088.8	-89.8
4	0.567	1.091	0.77	-1210.8	-121.9
5	0.587	0.990	n/a	-1321.6	-110.9
6	0.605	0.945	n/a	-1375.7	-54.0

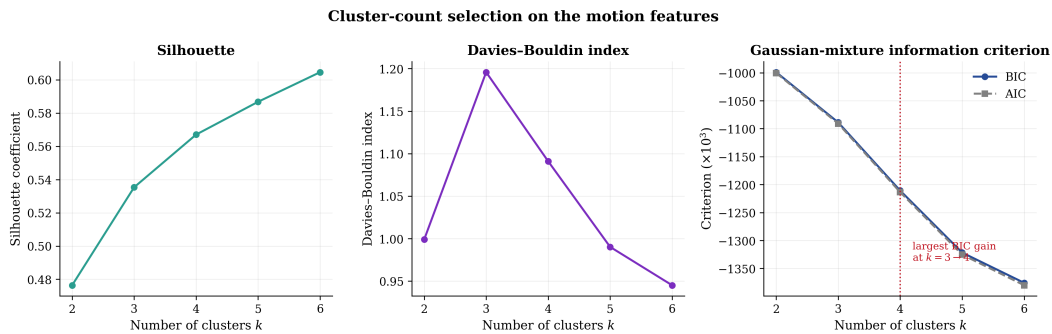


Figure 3.1: Cluster-count selection on the motion features. **(a)** Silhouette rises monotonically and **(b)** Davies–Bouldin keeps improving, so neither compactness index picks a k on its own. **(c)** The Gaussian-mixture BIC has its largest drop at $k = 3 \rightarrow 4$, but the $k = 4 \rightarrow 5$ drop is comparable; the marginal gain collapses only after $k = 5$. Four groups is retained on grounds of parsimony and physical interpretability (see text and Table 3.1).

Table 3.2 numbers the four clusters and, for each, gives its held-out share, its median motion, and the reasoning that maps it onto an operating regime. The regime labels used in the rest of this report are read off from these medians rather than assigned in advance, so that the naming follows from the data.

Table 3.2: The four clusters on the held-out window, numbered in duty-cycle order. Shares and medians are computed on ten-second windows; velocity and acceleration are in normalised units. The final column states why each cluster’s median motion identifies it with the named regime.

Cluster	Median vel.	Median accel.	Share (%)	Interpretation and justification
1	0.000	0.000	48.12	Zero speed and zero acceleration: the train is at rest, labelled <i>standing</i> .
2	0.305	0.041	14.34	Moderate speed with the largest positive median acceleration: the train is speeding up, labelled <i>accelerating</i> .
3	0.842	0.002	17.27	Highest median speed with near-zero acceleration: steady running, labelled <i>cruising</i> .
4	0.343	−0.033	20.26	Moderate speed with clearly negative acceleration: the train is slowing down, labelled <i>deceleration</i> .

The resulting profile matches a typical metro duty cycle. Standing dominates at about 48% of windows, with zero median speed and acceleration. Cruising has the highest median speed (0.84) and near-zero acceleration. Accelerating and deceleration sit at intermediate speeds and are separated by the sign of acceleration (+0.041 against −0.033). Deceleration accounts for about one window in five (20.3%), which sets the scale for the braking analysis in Chapter 5. Figure 3.2 shows the four motion profiles, and Figure 3.3 the velocity–acceleration cycle the train traces through them.

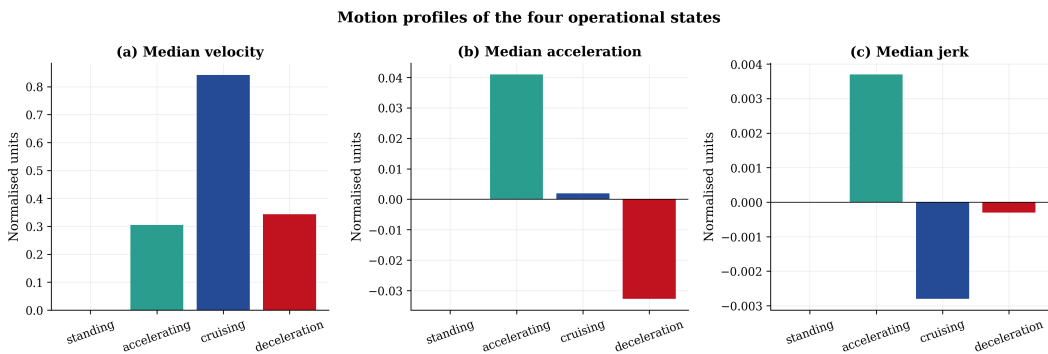


Figure 3.2: Motion profiles of the four states: (a) median velocity, (b) median acceleration, and (c) median jerk. The states are separated by speed and the sign and magnitude of acceleration; no pneumatic channel enters the definition.

3.2 Plausibility of four states

An earlier version of this analysis included brake-pressure sensors in the state definition. It systematically mislabelled a stationary train holding its brake at a station as deceleration: the brake was applied, so the pneumatic signal resembled braking even though the train was not

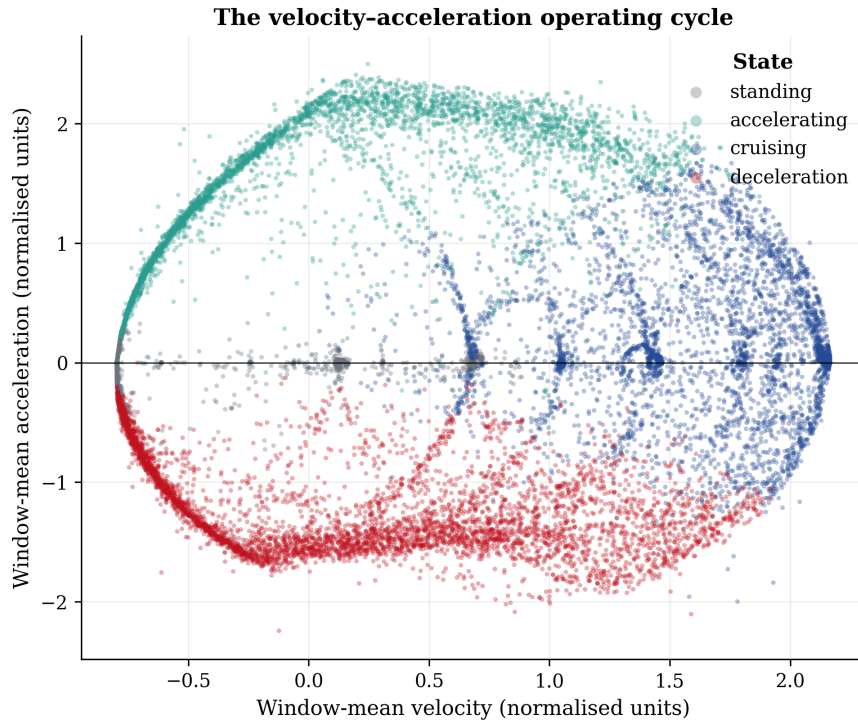


Figure 3.3: The velocity–acceleration cycle, coloured by state. The train moves anticlockwise: stop, accelerate, cruise, decelerate, stop. A minority of off-curve points do not fit the idealised cycle and motivate future work (Chapter 7).

moving. This is precisely the circularity that the feature-role split (Section 2.5) is designed to avoid, and it is the reason the state definition was restricted to motion. Table 3.3 shows how the earlier, pneumatics-contaminated “braking” label redistributes under the motion-only definition. Of the windows the old label called braking, about 105,000 are genuine deceleration, roughly 80,000 are in fact acceleration, and about 47,000 are a stationary train holding its brake. Rather than discarding this information whether a standing train is braked is retained as a sub-label (`standing_braked` versus `standing_idle`), but it no longer contaminates the deceleration regime, which is 99.9% genuine deceleration under the corrected definition.

Table 3.3: Reassignment of the old, pneumatics-contaminated “braking” label under the motion-only definition. Windows counted at the window level.

Motion-only state	Windows from old “braking”
accelerating	80,226
deceleration	104,908
cruising	519
standing	46,723

As a second check, the motion-only clustering was compared against a classifier trained on the full feature set. A Random Forest predicting the state labels from both the kinematic and the pneumatic window summaries reaches 0.994 accuracy on held-out windows, and its most important features are the kinematic ones: the mean, maximum, and minimum of

acceleration, and the mean and spread of speed. Pneumatic channels appear only much lower in the ranking. Even when the pneumatic signals are made available, then, the states remain separable by motion alone, which confirms that they are motion-defined regimes rather than an artefact of the pneumatic data.

3.3 Transitions and dwell

The states occur in the expected order (stop, accelerate, cruise, brake, stop), and the transition matrix is dominated by this cycle (Figure 3.4). Mean dwell times match the physical picture: standing about 54 s (with very high variance, since layovers and terminus stops are long and irregular), accelerating about 17 s, cruising about 21 s, and deceleration about 24 s. The state mix is stable from month to month (Figure 3.5), which indicates that the regimes are a property of how the train is operated rather than of any particular period in the recording.

Operational-state transition probabilities

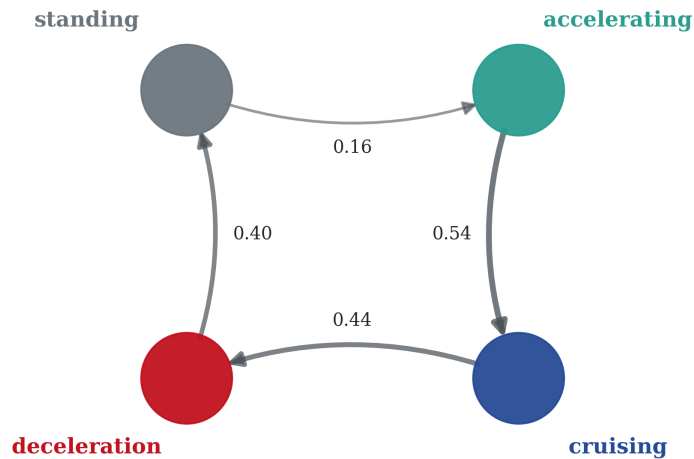


Figure 3.4: State transition diagram. The dominant path is the operating cycle stop $\rightarrow$ accelerate $\rightarrow$ cruise $\rightarrow$ decelerate $\rightarrow$ stop.

3.4 Robustness to window length

Because the ten-second window is a design choice, the clustering was re-run at 5, 10, 15, and 20 seconds as a read-only sensitivity check that left the canonical ten-second models untouched (Figure 3.6). The deceleration share is essentially flat across all four sizes, about 20% at each. The state *chatter* rate, the fraction of windows whose state differs from an adjacent window, rises with window length, from about 0.19 at 5 s to 0.54 at 20 s. Short windows are therefore

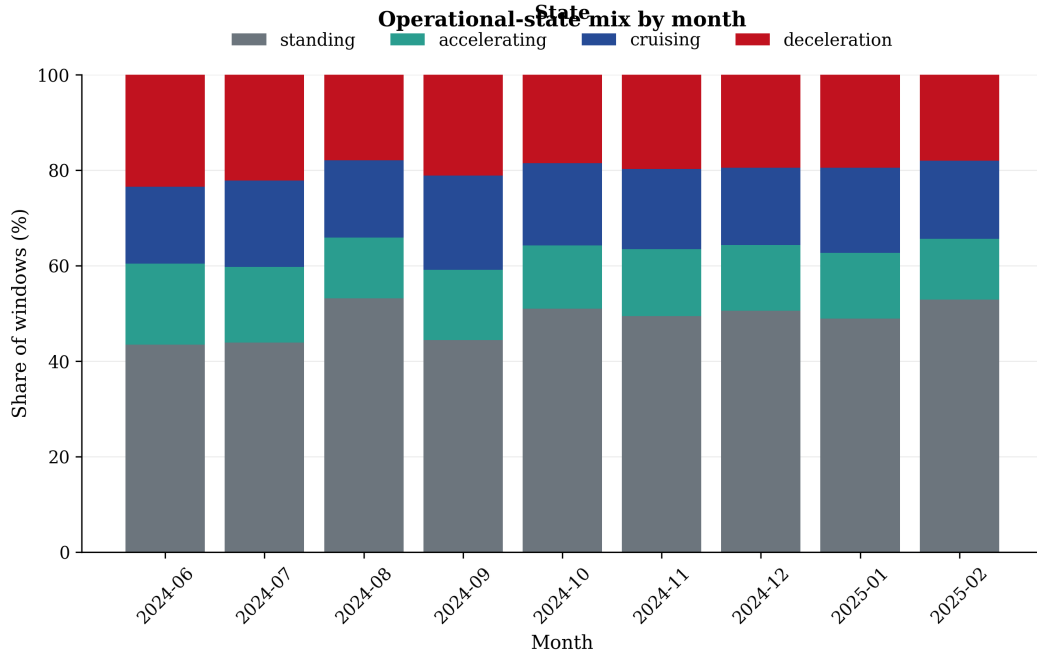


Figure 3.5: State mix by month. The proportions are stable across the recording, so the regimes reflect the operating pattern rather than a seasonal artefact.

not the source of label churn, and ten seconds is a sensible compromise between resolution and stability.

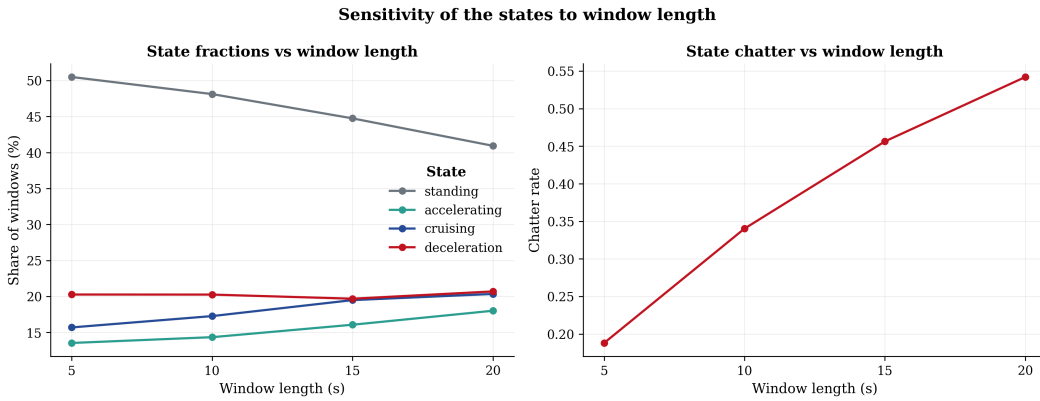


Figure 3.6: Window-length sensitivity. State fractions (left) are stable across 5–20s; the deceleration share stays near 20%. Chatter (right) rises with window length, so short windows are not the cause of label churn.

3.5 Pre-event state distribution

As an initial link to the maintenance record, the mix of operational states in the 24 to 72 hours before each documented failure or maintenance event was compared with the overall baseline. Across 62 event-by-window-size chi-square tests, 61 were significant at $p < 0.05$. This significance is, however, an artefact of sample size: with 6,000 to 24,000 windows per test,

even negligible differences reach significance. The effect size is small throughout: Cramér’s V does not exceed 0.07 and is typically about 0.018, an order of magnitude below the 0.1 value conventionally regarded as small.

The shift that does exist reflects the train standing idle in the depot before scheduled work. Ahead of planned revisions the standing fraction rises by roughly 45 to 52 percentage points. Restricting the test to moving windows, by dropping the standing state, reduces the effect size roughly sixfold, to a Cramér’s V of about 0.003, which is indistinguishable from zero. For the eight brake failures specifically there is no such idling, since failures occur in service, and no meaningful effect size either.

The test therefore confirms only that the pipeline registers when the train is parked before scheduled maintenance, which is an operational sanity check rather than evidence that pre-failure braking behaviour differs from normal operation. This is consistent with the near-null pre-failure braking signal reported in Chapter 6, and it motivates examining charging and idle windows with the auxiliary sensors, rather than deceleration, in future work.

4 Classification

The operational states of Chapter 3 were defined from motion alone. This chapter asks the leakage-free question that motivates the whole design: can the *air-system* sensors, which never touched that definition, recover the deceleration state on their own? If they can, deceleration leaves an independent fingerprint across the pneumatic system, which is the premise the later analysis rests on.

4.1 Setup

The target is the binary label “deceleration versus not” on ten-second windows. The motion channels that defined the target are excluded from the predictors entirely, in line with the role ledger (Section 2.5). Two predictor tiers are compared:

- **auxiliary only** (60 features): the channels classed *auxiliary* in the role ledger (Section 2.5): the health and support channels (main-reservoir and load pressure, load signal, energy-braking resistance, ambient temperature, and compressor state). They are termed auxiliary because they never carry the brake command: they report the state of the air system rather than the braking demand itself, so recovering deceleration from them alone is the strictest test of independent signal.
- **full** (180 features): the auxiliary channels together with the *actuation* channels (the brake command and its immediate pneumatic response), which, unlike the auxiliary channels, are driven directly by the braking demand.

Several transparent models are fitted so that the result does not hinge on one method’s quirks: a decision tree, LDA, QDA, logistic regression, and a random forest. All are trained on the training window and scored on the held-out window (Section 2.2). Because deceleration is the minority class (about 20% of windows), balanced accuracy and ROC-AUC are read in preference to raw accuracy.

4.2 Results

The headline numbers are in Table 4.2. The full-tier random forest reaches a ROC-AUC of 0.954 on held-out data, with balanced accuracy 0.869 and recall 0.956 for the deceleration class. Its precision is lower, 0.528: the model catches almost all decelerations but also flags a fair number of non-decelerations, which is the expected trade-off when recall is high on a minority class and is the reason balanced accuracy and AUC are the yardsticks here rather than precision alone. LDA and the decision tree follow closely (AUC 0.924 and 0.910); the decision tree is the simplest model yet lands near the top, making it the most interpretable

strong option. QDA reaches 0.901 and logistic regression 0.908. Full per-model metrics for both tiers are collected in Appendix B.

The auxiliary-only tier result is the most interesting. With the brake command *withheld*, the random forest still reaches AUC 0.921. Deceleration therefore leaves a fingerprint across the independent health sensors (reservoir pressure, load pressure, and the rest) even without being told the brake was applied. Most of the recoverable signal lives in the auxiliary channels, not only in the actuation channels, which is what makes the pneumatic system a plausible place to look for wear.

Which channels carry the signal

The interpretable models agree on where the deceleration signal sits. The decision tree puts almost all of its weight on a single channel, pneumatic braking force, with spring-brake pressure and proportional-valve pressure filling in the lower splits (Table 4.1). LDA spreads its weight more evenly but ranks the same subsystem first: the per-wagon pneumatic-braking-force and proportional-valve-pressure copies dominate its discriminant, with reservoir pressure the strongest auxiliary contributor. QDA exposes no single loading vector, but it reaches comparable AUC on the same channel pool, so all three lean on the same part of the pneumatic system. The channels that matter are the ones that actuate and support braking, and none of them entered the motion-only state definition.

Table 4.1: Top air-system channels for the full-tier classifiers. The decision-tree column is the split importance; the LDA column is the standardised discriminant weight ($|\text{coef}|$). Per-wagon copies are collapsed to their channel family.

Channel (family)	Decision tree	LDA $ \text{coef} $
pneumatic braking force	0.69	high
spring-brake pressure	0.14	low
proportional-valve pressure	0.11	highest
energy-braking resistance	0.05	low
main-reservoir pressure	0.00	moderate

Table 4.2: Deceleration classification on the held-out window. The full-tier random forest leads; the auxiliary-only random forest, with the brake command withheld, is nearly as strong. AUC is measured against 0.5 (coin flip); balanced accuracy against per-class chance.

Tier	Model	ROC-AUC	Balanced acc.	F_1 (decel.)
full	random forest	0.954	0.869	0.680
full	LDA	0.924	0.726	0.610
full	decision tree	0.910	0.825	0.609
full	QDA	0.901	0.746	0.602
auxiliary only	random forest	0.921	0.831	0.620

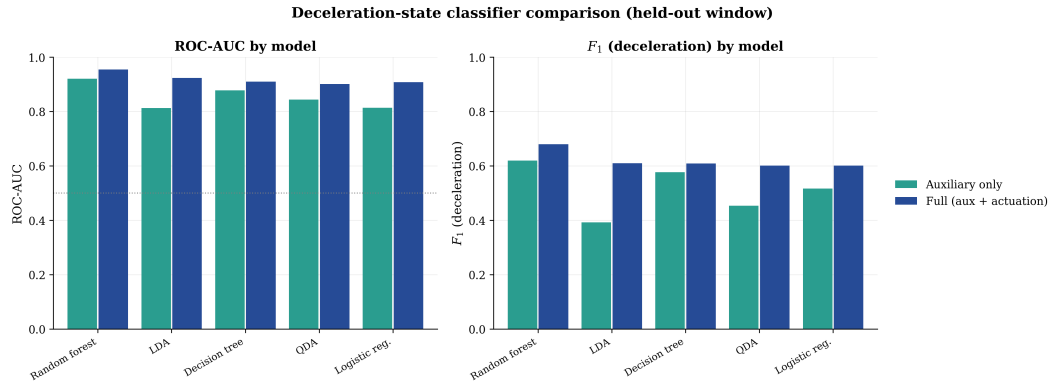


Figure 4.1: Model comparison for deceleration recovery (held-out AUC and F_1). The full-tier random forest leads; the auxiliary-only tier, without the brake command, remains close.

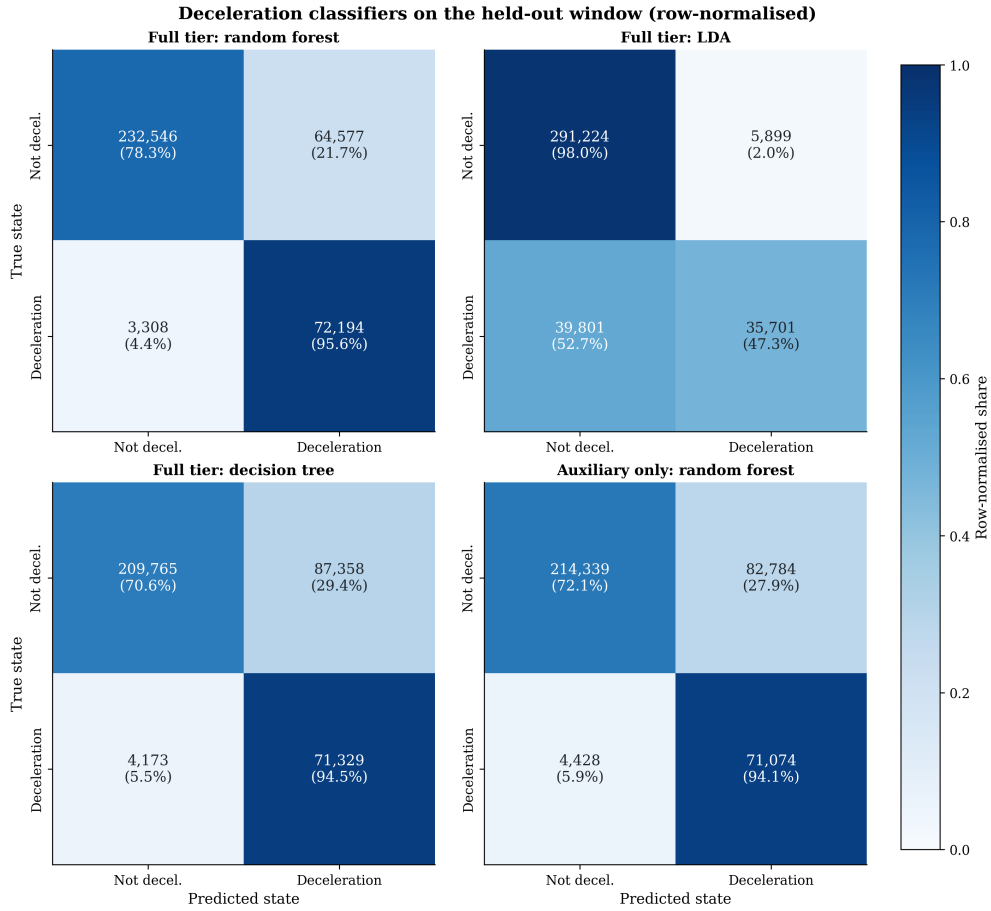


Figure 4.2: The four most informative confusion matrices on the held-out window (row-normalised). The full-tier random forest catches almost every deceleration at the cost of some false positives; LDA is the opposite, precise but conservative; the decision tree sits between them; and the auxiliary-only random forest, with the brake command withheld, keeps the same shape, showing the signal survives without actuation channels.

5 Analysis of braking events

5.1 Events

A deceleration event is one continuous stretch of the deceleration state; the training window yields about 126,700, of which 99.9% are genuine decelerations once the held-brake contamination was removed (Section 3.2). The mean event lasts 23.4s and the median 20s (95th percentile 30s): mostly ordinary station stops, with a thin tail of longer ones.

5.2 Classes or continuum?

The original plan anticipated discrete braking classes: light, hard, emergency. The data was allowed to decide instead. Clustering the intensity features and reading BIC, silhouette, and resampling stability together turns up no natural class count: the BIC keeps improving as clusters are added rather than settling at a minimum, and the silhouette never singles out a k (Figure 5.1). Stability does rise again at $k = 5$, which looks stable at a glance but survives only by peeling off a cluster holding a few hundredths of a percent of the events, unsupported by BIC or silhouette: the bump is an artefact of one tiny outlier group, not a regime. The intensity histogram seems to pull the other way, with a dominant mild mode and a broader hump at firmer braking (Figure 5.2), so on its own the density looks bimodal. But the humps overlap without a clean gap, and the resampling scrambles any boundary drawn between them, so the clustering never confirms the visual split. That tension is left as an open question; on the available evidence the report treats intensity as a continuum, varying by degree rather than by kind. No light/hard/emergency labels are introduced, then. Any such split is an *operational threshold* to be set with domain input, not a structure the data volunteers.

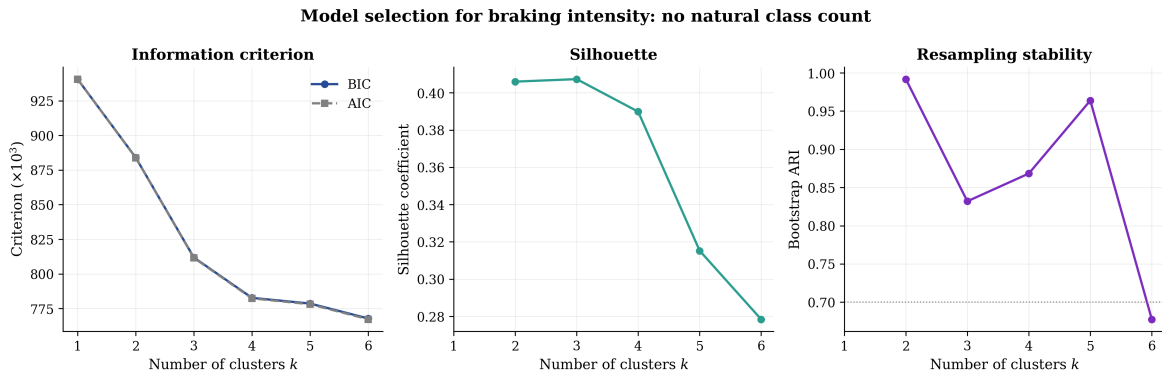


Figure 5.1: Model selection for braking intensity. BIC keeps improving with more groups and the silhouette falls away, so there is no natural number of intensity classes. The resampling stability rises again at $k = 5$, but that solution only splits off a negligibly small cluster and no other criterion supports it, so the bump carries no interpretable meaning: intensity is a continuum.

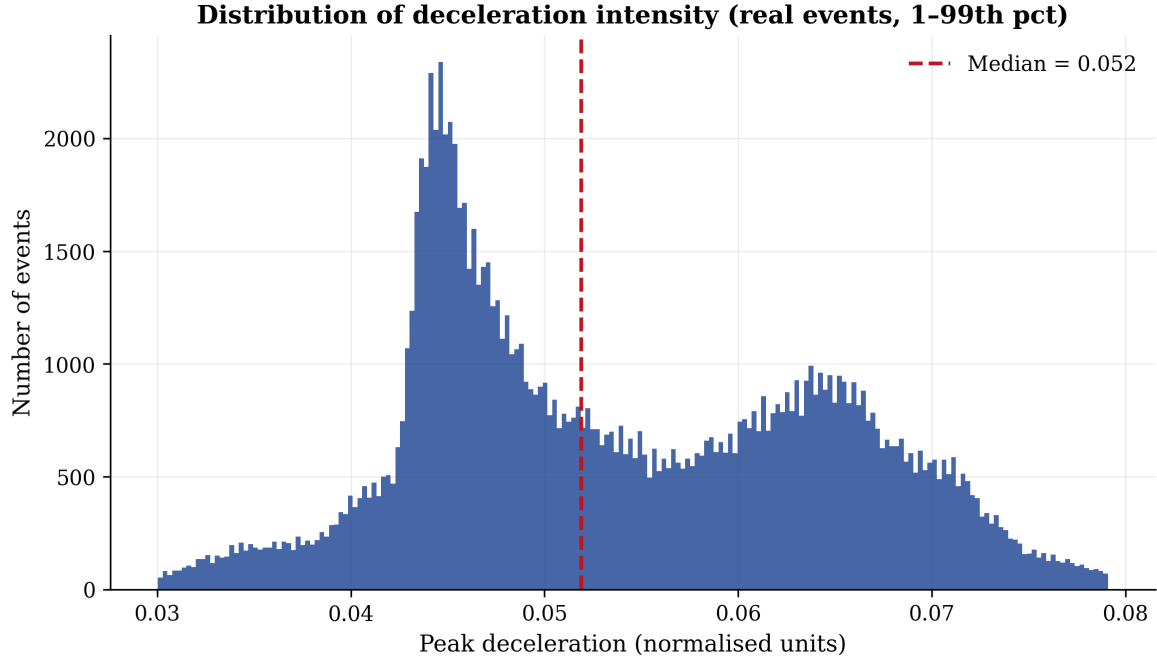


Figure 5.2: Distribution of deceleration intensity. A dominant mild mode and a broader firmer-braking hump overlap without a clean gap; the cluster tests find no stable boundary between them, so intensity is treated as a continuum rather than as discrete classes.

5.3 A deceleration-energy axis

To compare stops on a physical scale, a specific deceleration energy is derived, $\frac{1}{2}(v_0^2 - v_1^2)$, the kinetic energy dissipated per stop per unit mass. Being kinematic, it carries the role *velocity* and serves only as a target, never a predictor. Adding it to the intensity features does not create classes (the continuum of Section 5.2 is unchanged) but gives a common wear axis, putting every stop on the same energy-dissipated scale. The air-system sensors explain about 85% of a stop’s specific energy out-of-sample (full tier, random forest), so the pneumatic record tracks dissipated energy closely even without seeing the speed.

5.4 Intensity regression

The central cross-modal test predicts deceleration *magnitude* from the non-speed sensors on held-out events, against a guess-the-average baseline ($R^2 \approx 0$ by construction). A random forest on the full tier reaches $R^2 \approx 0.43$ for peak and 0.57 for mean deceleration; a linear model manages only 0.24 and 0.20, so most of the signal is non-linear structure a straight fit misses. These are moderate results: the pneumatic sensors carry real but partial information about how hard the train braked, not a near-perfect reconstruction. The scatter of predicted against actual intensity (Figure 5.3) is positively associated but sits in a narrower band than the targets: the gentlest stops are pushed up and the hardest pulled down, flattening the cloud against the diagonal. This is regression toward the average, where a squared-error model

retreats when the inputs leave real variance unexplained, and the per-event pneumatic signal is that noisy. Two other explanations were checked and rejected. Not sensor saturation: the actuation pressures stay well inside range on hard stops, with only one to two percent of events near a ceiling. Not the ten-second windowing either: adding event duration to the predictors moves R^2 by less than a percentage point. The model did matter. A gradient-boosted tree scores 0.47 on mean deceleration in a tighter band; the random forest reaches 0.57 and widens the spread by about a fifth, since deep independent trees reach more extreme conditional means than a shallow boosted ensemble. Peak deceleration holds at 0.43 either way, so its residual band is the limit of what these sensors carry, not a model shortcoming. Both panels of Figure 5.3 use the random forest. A Lasso fit with its penalty cross-validated on the training events keeps 19 of the 22 air-system features ($\alpha \approx 4 \times 10^{-6}$), a mix of actuation and auxiliary channels spread across subsystems rather than concentrated in the brake command. The near-minimal penalty and broad selection say the intensity signal is distributed, not localised, matching the auxiliary-only classification result of Chapter 4.

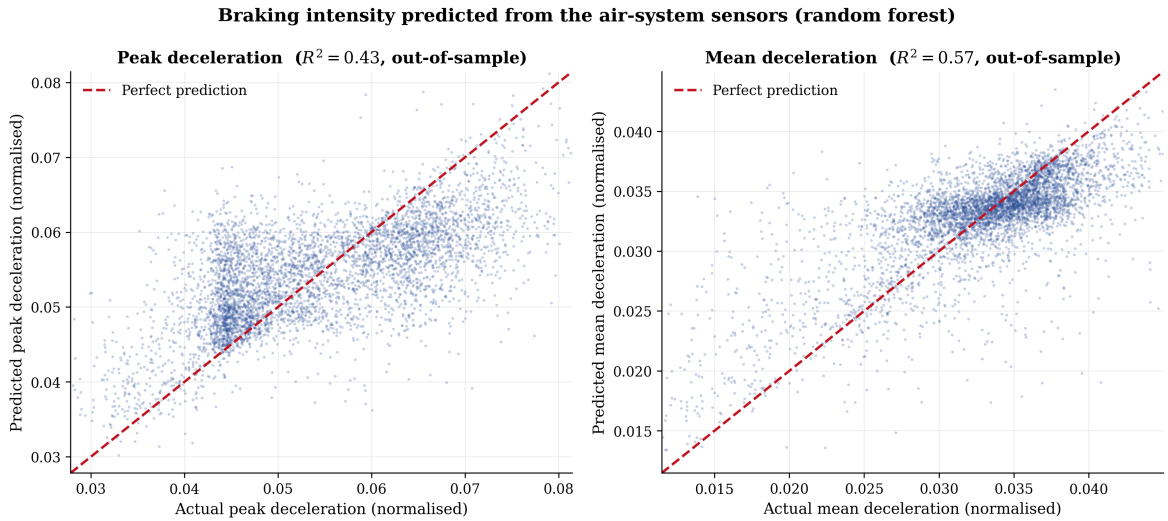


Figure 5.3: Predicted versus actual deceleration intensity on held-out events (full tier, random forest). The association is positive, but predictions sit in a narrower band than the targets: the gentlest stops are over-predicted and the hardest under-predicted, the regression-toward-the-average that remains once the pneumatic sensors have given up their partial signal.

6 Connection to failure events and maintenance

The preceding chapters established that motion defines four operational states, that the air-system sensors independently recover the deceleration state, and that they explain a moderate share of braking intensity. This chapter turns to the question behind predictive maintenance: does braking behaviour change in the days before a documented brake-system failure? The evidence points to a near-null answer, reported here as a genuine negative rather than qualified away.

6.1 Setup

The training window contains 10 documented failures (8 brake-system) and 11 maintenance events. Deceleration events in the seven days before any brake failure form a pre-failure group (about 23,000 events), compared against a baseline more than 30 days from any failure (about 44,000). Ten event-level features are tested, spanning the kinematic intensity measures and the auxiliary health channels.

6.2 Distributional test

Each feature is compared between the pre-failure and baseline groups with a Mann–Whitney U test [3], a rank-based test that asks whether values from one group tend to exceed the other without assuming normality: it ranks the pooled values and counts, over every cross-group pair, how often the pre-failure value is larger. Under the null that count is about $n_1 n_2 / 2$, and the p -value measures the departure from it. With ten features tested, the p -values are Bonferroni-corrected: at $m = 10$ and $\alpha = 0.05$ a feature counts as significant only if its raw p falls below $\alpha/m = 0.005$, capping the family-wide false-positive chance at α . With more than 23,000 pooled events almost every feature clears that bar, which means little here: at this sample size even trivial differences drive p to zero. What is read instead is the effect size, Cliff’s delta [1], where 0 is no separation and ± 1 total, with $|\delta| < 0.15$ treated as negligible.

The result is near-null (Table 6.1). The kinematic and braking-command features (peak and mean deceleration, deceleration energy, brake-cylinder and pneumatic-braking-force channels) all sit at $|\delta| < 0.06$, effectively no separation despite significant p -values. Only energy-braking-resistance clears the threshold, at $|\delta| \approx 0.20$; the next largest, load pressure at ≈ 0.10 , stays negligible. Even that effect is small and rests on only eight brake failures, so it cannot support a prediction claim on its own. It does hint at *where* any signal might live, in the auxiliary channels rather than the deceleration kinematics, a point the conclusion returns to.

Table 6.1: Pre-failure versus baseline effect sizes (Cliff’s delta), top 7 $|\delta|$ values. The braking-kinematic features are negligible; only one auxiliary channel (energy-braking-resistance) rises above $|\delta| = 0.15$, and even that is small and rests on eight brake failures. Negative δ means lower values before failure.

Feature	Cliff’s δ	Above negligible ($ \delta > 0.15$)?
energy-braking-resistance (mean)	−0.201	yes
load pressure (mean)	−0.103	no
jerk RMS	+0.054	no
brake-cylinder pressure (integral)	−0.040	no
deceleration energy (specific)	−0.039	no
pneumatic braking force (mean)	−0.032	no
peak deceleration	−0.005	no

6.3 Change-point detection

As an independent check, a CUSUM (cumulative-sum) change-point detector [4] was run on the strongest signal from the previous section, the weekly mean of energy-braking-resistance (Figure 6.1). One design choice matters here and applies to this figure alone. The raw weekly proxy climbs almost monotonically across the year, which would only make sense if brake wear never reversed. Assuming the brakes are serviced, the proxy is reset to its post-service level at every maintenance date and only the within-interval drift is read. This is an assumption, not a fact: the log records that a stop happened, not that the brakes were touched, so the reset may not hold at every stop. It is adopted because a proxy that only ever rises is not sensible to run a change-point test on.

CUSUM keeps a running total of each week’s deviation from the established mean: while the process is stable the positive and negative deviations cancel and the sum stays near zero, but once the level shifts they share a sign and the sum climbs until it crosses a threshold and flags a change-point. A persistent drift thus surfaces where one-off noise does not. The threshold is four standard deviations, tuned for about one false alarm every six months. On the reset signal it fires four times, only one within seven days of a documented failure. There is no reliable alignment between change-points and failures during the deceleration phase, consistent with the near-null effect sizes of Section 6.2.

6.4 Energy and the maintenance calendar

The deceleration-energy axis of Section 5.3 was examined descriptively against the maintenance calendar (Figure 6.2). Energy totals track how long an interval ran and how much the train moved, following interval length more than failure risk. A CUSUM on weekly summed energy fires three times, again only one within seven days of a failure. As an accounting check, about 96% of all real-deceleration energy falls inside the between-maintenance intervals, so almost no braking is missed.

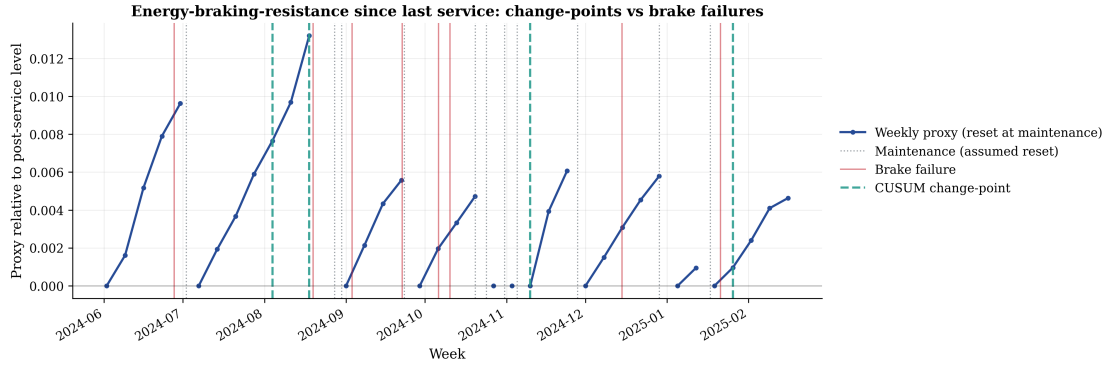


Figure 6.1: CUSUM change-points on the weekly mean of energy-braking-resistance, the auxiliary channel with the largest pre-failure effect size, reset to its post-service level at each maintenance date (dotted) under the assumption that the brakes are serviced there. Within each between-maintenance interval the proxy drifts up from zero; four change-points fire and only one falls within seven days of a documented failure, so even the strongest proxy does not track failures during the deceleration phase.

One pattern in the same figure is worth flagging, with care. Every between-maintenance interval containing a brake failure ran longer than about three weeks, and every shorter one was failure-free: the groups separate cleanly at roughly 21 days. On the service-at-each-stop reading of Section 6.3, a longer unmaintained stretch gives wear more time to build, which would make a cadence under three weeks a sensible target. Two cautions hold. A longer interval also spans more calendar days, so it has more chance to contain any event at all, confounding length with exposure; and the pattern rests on ten intervals and eight brake failures. It is an idea worth testing on more data, not a result.

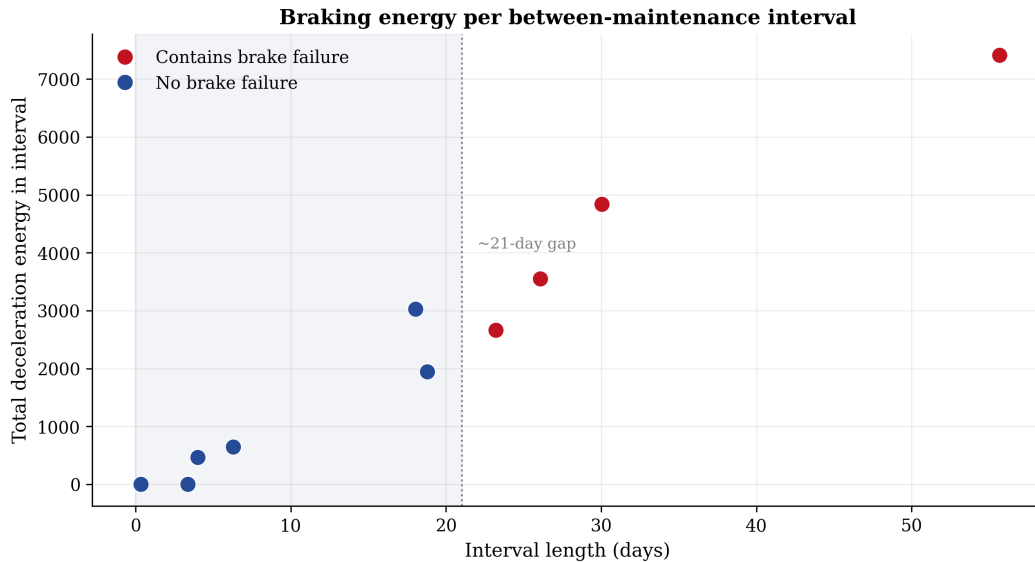


Figure 6.2: Deceleration energy per between-maintenance interval. Totals rise with interval length. Every interval that contained a brake failure (red) ran longer than about three weeks; every shorter interval was failure-free. The split is suggestive but confounded with exposure time and rests on ten intervals.

7 Conclusion

Motion alone splits a year of running into four operational states (standing, accelerating, cruising, deceleration), and the air-system sensors, held out of that definition, recover the deceleration state on their own: a random forest reaches ROC-AUC 0.95 on held-out data, and 0.92 even with the brake command withheld. Deceleration therefore leaves an independent fingerprint across the pneumatic system. Its intensity is a continuum rather than a set of classes, and the same sensors explain a moderate share of it out-of-sample (peak $R^2 \approx 0.43$, mean 0.57) and about 85% of a stop’s dissipated energy. The link to failures is an honest negative: neither the effect-size tests, the change-point detector on the strongest channel, nor the energy-versus-maintenance view separates the pre-failure period from baseline at only eight brake failures. Two soft hints survive, a small residual effect in the auxiliary channels rather than the braking kinematics, and a clean split in which every maintenance gap longer than about three weeks coincided with a failure. Both point to where a future search might look; neither is a detector.

The one open method question from Chapter 5 is now settled. Cross-validating the Lasso penalty (five-fold, on the training events only, so no held-out data leaks into the choice) selects a small penalty ($\alpha \approx 4 \times 10^{-6}$) and keeps 19 of the 22 air-system channels. The intensity signal is spread across subsystems, not carried by the brake command alone.

Four directions follow, three concrete and one broad. First, the off-curve outliers in the velocity–acceleration cycle (Section 3.1) were never characterised and may flag unusual operating conditions. Second, the per-wagon sensor copies were collapsed rather than compared; differences between wagons could localise where wear develops. Third, the pre-failure tests should be repeated on the standing and cruising states, where charging and idle behaviour (compressor duty cycle, reservoir recharge, spring-brake pressure at rest) is most visible and where brake faults are likeliest to actually show. Underlying all three is the reason none of them is guaranteed to pay off: this study found no dependable indicator of physical wear, and a dataset with many more documented failures is what would let any of these turn a hint into evidence.

The full pipeline, from raw recordings to the figures and tables in this report, is available on GitHub so the analysis can be reproduced end to end.¹

¹https://github.com/maxscheiblaue/Interdisciplinary_Project

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A Feature roles

This appendix lists the assignment of MetroAT channels to the feature roles introduced in Section 2.5. Only *actuation* and *auxiliary* channels are eligible as predictors; *velocity* channels define the targets, and *other* channels are used only for the train/held-out split, time alignment, and validation. Pneumatic channels are replicated per wagon (for example ..._BOGIE1, ..._BOGIE2); each entry below stands for the whole family of per-wagon copies, which are collapsed by principal component analysis (Figure 2.1) where a compact input is required. The exhaustive, machine-readable assignment is stored in `feature_roles.csv`.

Table A.1: Assignment of MetroAT channels to feature roles.

Role	Predictor	Channels
velocity	no	train speed (TRAIN_SPEED_ACTUAL) and the derived acceleration, jerk, velocity-change-rate, and deceleration-energy features
actuation	yes	brake-cylinder pressure, proportional-valve pressure (*_PROPORTIONAL_VALVE_PRESSURE_*), spring-brake pressure, and pneumatic braking force (*_PNEUMATIC BRAKING_FORCE_*)
auxiliary	yes	main-reservoir pressure, load pressure, load signal, energy-braking-resistance, ambient temperature, and compressor-running
other	no	timestamps, operating-mode flags (TRAIN_*_MODE), and the failure and maintenance markers (TRAIN_IS_IN_FAILURE, TRAIN_IS_IN_MAINTENANCE, TRAIN_FAILURE_TYPE, TRAIN_MAINTENANCE_TYPE)

A.1 Kinematic features

The operational states of Chapter 3 are clustered from kinematic (*velocity*-role) features alone. These rest on a single raw channel from the dataset, the train speed, from which acceleration, its absolute rate, and jerk are derived on the time-sorted 1 Hz stream. All derivatives are Δt -aware (they use the actual time gap between consecutive samples) and are set to zero across gaps longer than 2 s, so that out-of-service breaks do not create spurious spikes. Table A.2 lists the four base signals with their origin and, where derived, their formula. Here v_t is the normalised speed at sample t and Δt the interval to the previous sample (nominally 1 s).

Each base signal is reduced to per-window summaries over the ten-second window. The clustering uses twelve window features: the mean and standard deviation of all four signals (eight features), together with the minimum and maximum of speed and of acceleration (four features). The event-level kinematic descriptors used for braking intensity in Chapter 5 (peak and mean deceleration, Δv , jerk peak and RMS, and the two deceleration-energy axes) are derived from the same speed channel and are defined where they are first used.

Table A.2: The kinematic base signals: origin (raw dataset channel versus crafted) and, for the crafted signals, their definition. All values are in the normalised units of the dataset; rates are per second.

Signal	Origin	Definition
train speed v	dataset	raw channel TRAIN_SPEED_ACTUAL, min-max normalised to $[0, 1]$
acceleration a	crafted	$a_t = \frac{v_t - v_{t-1}}{\Delta t}$
velocity-change rate	crafted	$ a_t = \left \frac{v_t - v_{t-1}}{\Delta t} \right $, the unsigned acceleration
jerk j	crafted	three-sample rolling mean of $\frac{a_t - a_{t-1}}{\Delta t}$ (the smoothed rate of change of acceleration)

B Full classifier metrics

Table B.1 reports every model scored in Chapter 4 on the held-out window, for both predictor tiers. The main text quotes ROC-AUC, balanced accuracy, and F_1 ; the precision and recall columns are added here to make the precision/recall trade-off explicit. Deceleration is the minority class (about 20% of windows), so accuracy alone is misleading and balanced accuracy and ROC-AUC are the primary yardsticks. All numbers are read directly from `state_classification_metrics.csv`.

Table B.1: Deceleration classifiers on the held-out window, all models and both tiers, ordered by ROC-AUC within tier. Precision, recall, and F_1 are for the deceleration class.

Tier	Model	Acc.	Bal. acc.	Prec.	Recall	F_1	ROC-AUC
full	random forest	0.818	0.869	0.528	0.956	0.680	0.954
full	LDA	0.877	0.726	0.858	0.473	0.610	0.924
full	decision tree	0.754	0.825	0.449	0.945	0.609	0.910
full	logistic reg.	0.750	0.817	0.444	0.929	0.601	0.908
full	QDA	0.844	0.746	0.622	0.583	0.602	0.901
aux. only	random forest	0.766	0.831	0.462	0.941	0.620	0.921
aux. only	decision tree	0.731	0.796	0.424	0.906	0.577	0.878
aux. only	QDA	0.521	0.693	0.295	0.983	0.454	0.844
aux. only	logistic reg.	0.719	0.728	0.397	0.744	0.517	0.814
aux. only	LDA	0.820	0.621	0.621	0.287	0.392	0.813